## LAB : Spark

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

```
bmscecse@bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ spark-shell
26/05/04 16:13:42 WARN Utils: Your hostname, bmscecse-HP-Pro-Tower-400-G9-PCI-Desktop-PC resolves to a loopback address: 1
26/05/04 16:13:42 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/04 16:13:48 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
Spark context Web UI available at http://10.124.3.47:4040
Spark context available as 'sc' (master = local[*], app id = local-1777891430182).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.4
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 11.0.26)
Type in expressions to have them evaluated.
Type :help for more information.

scala> val textFile = sc.textFile("/home/bmscecse/wc.txt")
textFile: org.apache.spark.rdd.RDD[String] = /home/bmscecse/wc.txt MapPartitionsRDD[1] at textFile at <console>:23

scala> val counts = textFile.flatMap(line => line.split(" ")).map(word => (word,1)).reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:23

scala> import scala.collection.immutable.ListMap
import scala.collection.immutable.ListMap

scala> val sorted = ListMap(counts.collect().sortWith(_._2 > _._2): _*)
sorted: scala.collection.immutable.ListMap[String,Int] = ListMap(hello -> 6, spark -> 6, big -> 2, data -> 2, world -> 2)

scala> for((k,v) <- sorted) { if(v > 4) println(k + "," + v) }
hello,6
spark,6

scala> 
```